



Quentin Le Houérou

PhD student
Paris-Est Créteil University (UPEC)
LACL

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Education

- **École normale supérieure de Lyon** 2019-2020
Bachelor's degree (L3) in Theoretical Computer Science
- **University of Lyon 1** 2019-2020
Bachelor's degree (L3) in Mathematics
- **École normale supérieure de Lyon** 2020-2021
Master's degree (M1) in Theoretical Computer Science
- **École normale supérieure de Lyon** 2021-2022
Preparation for the Agrégation in Mathematics (competitive teaching exam) Rank : 18th
- **Université Paris Cité** 2022-2023
Master's degree (M2), Mathematical Logic and Foundations of Computer Science (LMFI)
- **Université Paris-Est-Créteil-Val-de-Marne** 2023-present
PhD, "Metamathematics of Ramsey-like theorems" Supervisors: Ludovic Levy Patey & Julien Cervelle

Research Internships

- **Computability-theoretic analysis of the Rainbow Ramsey theorem** 2020
Supervisor: Ludovic Patey Institut Camille Jordan, Lyon
 - Computability theory
 - Reverse Mathematics
- **Gerbes et Torsors in homotopy type theory** 2021
Supervisor: Thierry Coquand University of Gothenburg, Sweden (Remote)
 - Homotopy type theory
 - Proof assistant: Agda
- **Conservation of Ramsey's theorem for pairs** 2023
Supervisors: Ludovic Levy Patey & Keita Yokoyama Tohoku University, Japan
 - Proof theory
 - Reverse Mathematics

Publications

1. Quentin Le Houérou, Ludovic Levy Patey, and Keita Yokoyama. Π_4^0 conservation of ramsey's theorem for pairs. *Journal of the London Mathematical Society*, 113(1):e70419, 2026
2. Lorenzo Carlucci, Oriola Gjetaj, Quentin Le Houérou, and Ludovic Levy Patey. Ramsey-like theorems for the schreier barrier. *The Journal of Symbolic Logic*, page 1–30, 2025
3. Quentin Le Houérou, Ludovic Levy Patey, and Keita Yokoyama. Conservation of Ramsey's theorem for pairs and well-foundedness. *Trans. Amer. Math. Soc.*, 378(3):2157–2186, 2025
4. Quentin Le Houérou and Ludovic Levy Patey. Π_4^0 conservation of the ordered variable word theorem. *The Journal of Symbolic Logic*, page 1–16, 2025

Preprints

1. Quentin Le Houérou and Ludovic Patey. Largeness notions and polytime translation for $\forall\Sigma_3^0$ -consequences of RT_2^2 , 2026
2. Quentin Le Houérou and Ludovic Patey. Partition theorems for ketonen-solovay largeness. *arXiv preprint arXiv:2602.08778*, 2026
3. Quentin Le Houérou and Ludovic Patey. The reverse mathematics of bounded ramsey's theorem for pairs. *arXiv preprint arXiv:2509.03688*, 2025
4. Quentin Le Houérou and Ludovic Patey. Ramsey-like theorems for separable permutations, 2025
5. Quentin Le Houérou, Ludovic Levy Patey, and Ahmed Mimouni. The reverse mathematics of the pigeonhole hierarchy, 2024

Teaching experiences

•Paris-Est Créteil University (UPEC)

2025-2026

- Tutorial: Sequences and series of functions.
- Practical session: Mobile programming.
- Tutorial: Data representation.

•Paris-Est Créteil University (UPEC)

2024-2025

- Practical session: functional programming
- Tutorial: Algebraic Grammar and Interpretation